6.5 Angles of Elevation and Depression

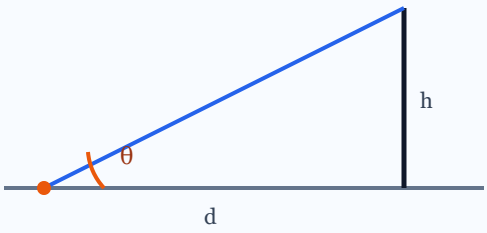
The **angle of elevation** is measured upward from a horizontal line of sight to an object above; the **angle of depression** is measured downward from horizontal to an object below. Because the horizontal lines at the observer and at the object are parallel, and the line of sight is a transversal between them, the angle of elevation from the lower point equals the angle of depression from the higher point — they are alternate interior angles. Every elevation/depression problem becomes a right triangle: the horizontal distance is one leg, the height difference is the other leg, and $\tan\theta = \dfrac{\text{height}}{\text{distance}}$ connects them (with sin or cos used when the hypotenuse — the actual line of sight — is involved instead). Some problems require an extra step, such as adding an observer's eye height, or subtracting two elevation readings to isolate the height of just part of an object (like a window). This worksheet models real situations with full multi-step right-triangle trigonometry.

Examples

Example 1: Basic elevation

From a point 40 ft from the base of a building, the angle of elevation to the top is 30°. Find the building's height.

Solution



Horizontal distance $= 40$; angle of elevation $= 30^\circ$; height is the unknown leg.

Identify the right triangle: horizontal distance $= 40$ (adjacent to θ), height $= h$ (opposite).

Apply TOA: $\tan 30^\circ = \dfrac{h}{40}$

Solve: $h = 40\tan 30^\circ = 40 \cdot \dfrac{1}{\sqrt{3}} = \dfrac{40\sqrt{3}}{3}$ ft.

Answer: $\dfrac{40\sqrt{3}}{3}$ ft.

Example 2: Depression to find distance

From the top of a 50 m cliff, the angle of depression to a boat is $25°$. Find the horizontal distance to the boat.

Solution

The angle of depression from the cliff top equals the angle of elevation from the boat (alternate interior angles, parallel horizontals) $= 25°$.

In the right triangle: opposite (cliff height) $= 50$, adjacent (distance) $= d$.

Apply TOA: $\tan 25° = \dfrac{50}{d}$

Solve: $d = \dfrac{50}{\tan 25°}$ m.

Answer: $d = \dfrac{50}{\tan 25°}$ m.

Example 3: Elevation of 45°

From 20 m away, the angle of elevation to the top of a pole is $45°$. Find the height.

Solution

Identify the right triangle: horizontal distance $= 20$ (adjacent to θ), height $= h$ (opposite).

Apply TOA: $\tan 45° = \dfrac{h}{20}$

Since $\tan 45° = 1$: $h = 20 \times 1 = 20$ m.

Answer: 20 m (at $45°$, height always equals distance).

Example 4: Prove the equal-angles relationship

Prove that the angle of elevation from a ground point A to a point B above equals the angle of depression from B back to A.

Solution

The horizontal line at A and the horizontal line at B are both horizontal, so they are parallel.

Segment $\overline{AB}$ (the line of sight) is a transversal cutting these two parallel horizontal lines.

The angle of elevation at A (above the horizontal at A) and the angle of depression at B (below the horizontal at B) are alternate interior angles formed by this transversal.

Conclusion: alternate interior angles of parallel lines are congruent, so the angle of elevation from A equals the angle of depression from B.

Example 5: Tree height at 60°

From 15 ft away, the angle of elevation to a tree's top is $60°$. Find the height.

Solution

Identify the right triangle: horizontal distance $= 15$ (adjacent to θ), height $= h$ (opposite).

Apply TOA: $\tan 60° = \dfrac{h}{15}$

$h = 15 \tan 60° = 15\sqrt{3}$ ft.

Answer: $15\sqrt{3}$ ft.

Example 6: Find the angle from a ramp

A ramp rises 4 ft over a horizontal run of 12 ft. Find the angle of elevation.

Solution

Identify the legs of the right triangle: rise (opposite) $= 4$, run (adjacent) $= 12$.

Apply TOA: $\tan \theta = \dfrac{4}{12} = \dfrac{1}{3}$

Solve with the inverse tangent, since the angle itself — not a side — is unknown:

$\theta = \tan^{-1}(\frac{1}{3})$.

Answer: $\theta = \tan^{-1}(\frac{1}{3})$.

Example 7: Depression from a lighthouse

From a lighthouse 80 ft tall, the angle of depression to a ship is $20°$. Express the ship's distance from the base.

Solution

By the equal-angles relationship, the angle of elevation from the ship is also $20°$.

Apply TOA: $\tan 20° = \dfrac{80}{d}$

Solve: $d = \dfrac{80}{\tan 20°}$ ft.

Answer: $d = \dfrac{80}{\tan 20°}$ ft.

Example 8: Window height (subtracting two elevations)

From 25 m away, the angle of elevation to the top of a window is 50°, and to the bottom of the window is 30°. Find the window's height.

Solution

Find the height to the top: $h_{\text{top}} = 25 \tan 50°$.

Find the height to the bottom: $h_{\text{bottom}} = 25 \tan 30°$.

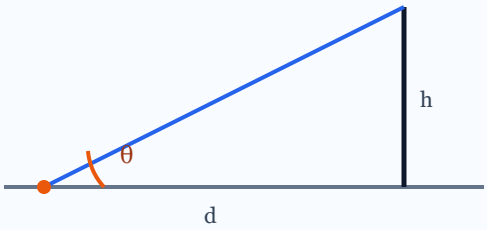
The window's height is the difference between these two heights: $h_{\text{top}} - h_{\text{bottom}} = 25 \tan 50° - 25 \tan 30°$.

Answer: $25(\tan 50° - \tan 30°)$ m.

Example 9: Ladder at an angle

A ladder leans at a 70° angle of elevation from the ground, with its base 3 ft from the wall. Find how high it reaches on the wall.

Solution



The ladder is the hypotenuse; the wall height is the opposite leg.

Identify the right triangle: horizontal base distance $= 3$ (adjacent to θ), height on the wall $= h$ (opposite).

Apply TOA: $\tan 70° = \dfrac{h}{3}$

$h = 3 \tan 70°$ ft.

Answer: $3 \tan 70°$ ft.

Example 10: Compound problem with eye height

A person whose eyes are 5 ft above the ground stands 60 ft from a tower. The angle of elevation from her eyes to the top of the tower is $35°$. Find the tower's total height.

Solution

Find the rise above eye level using the right triangle from eye level: $\tan 35° = \dfrac{x}{60}$

$x = 60 \tan 35°$ (height above eye level).

Add the eye height, since the sightline starts 5 ft off the ground, not at ground level: total $= 60 \tan 35° + 5$.

Answer: $(60 \tan 35° + 5)$ ft.

NAME: _____ Geometry (FL B.E.S.T.)

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Elevation

From 30 ft away, elevation 45°. Find the building height.

Answer: **Verification:**

Problem 2: Elevation

From 10 m away, elevation 30°. Find the height (exact).

Answer: **Verification:**

Problem 3: Depression

From 60 ft up, depression 45°. Find the horizontal distance.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 4: Angle

Rise 5, run 12. Find the angle of elevation.

Answer: **Verification:**

Problem 5: Tree

From 20 ft away, elevation 60°. Find the tree height.

Answer: **Verification:**

Problem 6: Depression

Cliff 100 m, depression 30°. Find the distance to the boat.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 7: Ladder

Base 4 ft out, elevation 65°. Find the height on the wall.

Answer: **Verification:**

Problem 8: Equal angles

The angle of elevation from the ground to the top of a tower equals what angle measured from the top?

Answer: **Verification:**

Problem 9: Elevation

Distance 50, elevation 20°. Find the height.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 10: Ramp

Rise 3 ft, run 15 ft. Find the angle.

Answer: **Verification:**

Problem 11: Depression

Height 45, depression 15°. Find the distance.

Answer: **Verification:**

Problem 12: Elevation

From 8 m away, elevation 45°. Find the height.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 13: Challenge — window height

From 40 ft away: elevation to the top of a window is 50°, to the bottom is 25°. Find the window's height.

Answer: **Verification:**

Problem 14: Challenge — eye height compound

A person's eyes are 6 ft above the ground, standing 45 ft from a tower. The angle of elevation from her eyes to the top is 40°. Find the tower's total height.

Answer: **Verification:**

Problem 15: Challenge — two depressions

From the top of a 120 m bridge tower, the angle of depression to a boat is 22°, and to a buoy farther away is 14°. Express the distance between the boat and the buoy.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

✓ Answer Key

1. 30 ft
2. $\frac{10\sqrt{3}}{3}$ m
3. 60 ft
4. $\tan^{-1}\left(\frac{5}{12}\right)$
5. $20\sqrt{3}$ ft
6. $100\sqrt{3}$ m
7. $4\tan 65^\circ$ ft
8. The angle of depression from the top to the ground
9. $50\tan 20^\circ$
10. $\tan^{-1}\left(\frac{1}{5}\right)$
11. $\frac{45}{\tan 15^\circ}$
12. 8 m
13. $40(\tan 50^\circ - \tan 25^\circ)$ ft
14. $(45\tan 40^\circ + 6)$ ft
15. $\frac{120}{\tan 14^\circ} - \frac{120}{\tan 22^\circ}$ m

Great work — keep practising!